

Chapter 2 — Review of Literature

2.1 Review of Relevant Literature

Hybrid Learning as an integrated training design

Hybrid learning is generally understood as the intentional combination of digital and face-to-face learning within one programme. In the training context, the term does not describe a simple division of hours between an online module and a classroom session. It describes a design in which the activities are connected. A digital activity may introduce basic information, allow employees to prepare before a session, or make reference material available later. A face-to-face activity may be used for discussion, demonstration, practice, feedback, and the clarification of difficult issues. The important question is whether the learner can see how one activity supports the next.

Garrison and Kanuka (2004) describe the potential of blended learning as arising from the thoughtful integration of face-to-face and online learning. Graham (2006) likewise treats blended learning as a combination of delivery modes, teaching methods, and learning approaches. These sources are largely concerned with education, but their central design principle is relevant to employee training: digital and in-person components should complement one another rather than duplicate or contradict each other.

Evidence from education supports the broad proposition that purposeful blending can be associated with improved learning outcomes, although it cannot be treated as direct evidence for bank employees. Vo, Zhu and Diep (2017) synthesised 51 effect sizes from higher-education courses and reported a small positive average effect for blended learning compared with conventional classroom teaching ($g = 0.385$). A later meta-analysis by Yu, Yu, Li and Wang (2025), published online in 2023, included 133 empirical studies involving 18,464 student participants and reported an upper-medium effect on student learning performance ($g = 0.651$). Both reviews concern student learning and use varied educational designs. They therefore provide background support for the value of a thoughtfully integrated approach, not proof that the same effect will occur in Union Bank training.

Kumar and Pande (2017) examine technology-mediated learning for working professionals. Their work is useful because it does not view technology as the sole explanation for successful learning. It discusses institutional, facilitator, learner, and pedagogical conditions within a blended-learning ecosystem. This perspective is relevant to an employee training programme because an accessible platform will have limited value if employees cannot see the relevance of the material, obtain guidance, or connect it with later practice.

Mulaudzi (2021) studied blended teaching in workplace training in the South African transport sector. Although the study is a Master's dissertation rather than a journal article, it offers useful workplace context. It considers employees' experience of flexibility, digital access, time management, learning material, and practical barriers. The study does not provide a ready-made corporate scale for the present setting. It does, however, show why employee wording must consider work schedules and access conditions rather than relying only on student-oriented language.

Han and Ellis (2020) developed the Perceptions of the Blended Learning Environment Questionnaire. Their work is important for the structure of the Hybrid Learning construct because it treats a blended environment as more than a general positive or negative opinion. It considers how learners experience

the relationship between modes and the contribution of the online component. The scale was validated in higher education, so its student wording cannot be transferred without change to bank employees. Nevertheless, it supports the decision to measure integration, platform access, material availability, and the opportunity to revisit digital content as related aspects of one Hybrid Learning construct.

Sosnova, Hlianenko, Sosnova, Tsyna and Tsyna (2025) provide the guide-suggested conceptual anchor for the present study. Their study examines hybrid education methods and learning effectiveness through an experiment involving 64 higher-education students. The experimental condition combined online platforms, practice-oriented activity, problem-based learning, interactive methods, and project work. The authors report stronger learning outcomes in the hybrid group than in the comparison group. The study does not use a corporate sample, does not measure employee training effectiveness, and does not publish a questionnaire for use in the present study. Its value lies in the direct conceptual proposition: a carefully designed hybrid learning approach can be examined in relation to learning effectiveness.

The population difference is important. Students and employees do not learn under identical conditions. Employees may complete learning alongside customer responsibilities, operational schedules, targets, and organisational procedures. The present study therefore does not assume that results from a student experiment apply automatically to a bank. It uses Sosnova et al. to frame the broad Hybrid Learning–Effectiveness question and relies on corporate studies to establish workplace relevance.

Hybrid training and effectiveness in corporate settings

Kim (2022) provides the principal corporate paper for the study. The paper reports a South Korean automotive sales-training programme that shifted across three delivery methods: traditional training, fully digital training, and hybrid training. The research used 583 training instances from 2019 and 2020: 325 participants in the traditional programme, 140 in the fully digital programme, and 118 in the hybrid programme. The hybrid format included both online and offline learning, while the fully digital format depended more heavily on self-directed learning.

Kim develops the Content, Instructional Design, Programmed Learning, and Recommendation framework, referred to as CIP-R, to assess training effectiveness. The framework includes dimensions such as content, instructional design, personalised learning, perceived competence, and recommendation. The paper reports that effectiveness declined after the move from traditional to fully digital delivery and improved after the introduction of the hybrid format. It also identifies instructor engagement, training duration, personalised learning, confidence and competence, and job-related skill level as factors associated with the reported training outcome.

The finding should be interpreted carefully. Kim does not conclude that hybrid delivery is always superior to classroom delivery. In the study, the traditional programme had the highest estimated effectiveness mean, hybrid delivery was in the middle, and fully digital delivery had the lowest mean. The difference between hybrid and fully digital delivery was not statistically significant in the reported comparison. The important contribution for the present study is therefore not a claim that any hybrid programme is automatically best. It is the evidence that training mode, instructor involvement, practice, and the design of learning activities all matter when evaluating workplace training.

Kim's study informs three decisions in the present research. First, Training Effectiveness is treated as more than a satisfaction score. Second, Hybrid Learning is measured through the quality and integration of the employee's experience, rather than by merely asking whether an employee attended an online

session. Third, Trainer Competence is kept separate from Hybrid Learning because Kim's findings indicate that human and design factors matter alongside the delivery mode.

Uddin, Ahamed, Jakowan, Islam and Nahar (2026) offer additional corporate evidence. Their study collected data from 150 employees with prior blended-learning experience. It examines whether blended-learning effectiveness is related to employee performance through soft-skill development and knowledge acquisition. The study found that blended learning was positively associated with the two learning-related variables, while the direct relationship with employee performance was not significant after those pathways were considered.

Uddin et al. are useful because they show that work-related outcomes should not be assumed to arise from training delivery alone. Learning-related processes matter. However, the present study does not reproduce the Uddin model. Soft-skill development, knowledge acquisition, and employee performance would create a broader model and require additional evidence and items. Uddin et al. are therefore used as supporting literature for the wider workplace relevance of blended learning, not as a base paper or as a source for a complete questionnaire.

Trainer Competence in hybrid training

Trainer competence has particular importance in a hybrid programme because employees may need guidance across more than one learning mode. A trainer needs subject knowledge and clarity, but may also need to explain the sequence between online activities and face-to-face sessions, make appropriate use of digital tools, invite questions, and provide feedback. These functions can help employees understand why they are asked to complete a digital activity and how the activity connects with later work practice.

The role of the trainer should not be confused with the technical quality of the LMS. A platform may be reliable and easy to use, while a trainer may still fail to explain its relationship to the learning objective. Similarly, an expert trainer may struggle to provide support if learning materials are inaccessible. The present study therefore treats Hybrid Learning and Trainer Competence as separate antecedent constructs. This allows a clearer assessment of whether programme design and trainer support are each associated with engagement.

The selected base papers do not provide one open workplace study that tests Trainer Competence and Learner Engagement together in a hybrid-training model. The present study therefore treats these as separate, literature-informed constructs rather than claiming that one base paper supplies the complete pathway. Kim (2022) identifies instructor engagement, training design, participation, and practice as relevant to the effectiveness of corporate hybrid training. Mulaudzi (2021) provides workplace blended-learning context on access, facilitation, material use, and practical barriers. Ansari et al. (2023), in a university setting, show that facilitation across digital and face-to-face modes requires trainer competence, even though their population differs from bank employees.

These sources do not prove the present relationship hypotheses. They provide a clear basis for separating the trainer's support from the employee's own involvement. The Trainer Competence items therefore cover subject knowledge, clarity, digital facilitation, connection of activities, participation support, and feedback. Learner Engagement is measured separately through attention, effort, participation, involvement, clarification-seeking, and interest. This keeps the questionnaire focused and avoids treating a favourable view of the trainer as the same thing as active employee participation.

Learner Engagement

Learner engagement refers to active involvement in the learning process. It may include attention, persistence, participation, effort, interest, and seeking clarification. Engagement is different from mere attendance. An employee may be present in a session, logged into a platform, or recorded as having completed a module without making a serious effort to understand the content. For that reason, course completion is an incomplete indicator of learning.

Schaufeli, Bakker and Salanova (2006) describe engagement in terms of vigour, dedication, and absorption in the work context. The present study does not transfer their work-engagement scale directly to training. It uses the broader idea that engagement involves active effort and involvement. The questionnaire asks whether employees stayed attentive, made an effort to understand, participated in activities, felt involved, sought clarification, and remained interested in completing the selected programme.

Learner engagement is positioned between the antecedent constructs and Training Effectiveness for a practical reason. An integrated learning programme and competent trainer can create better conditions for participation, but employees still need to invest attention and effort. When employees engage with the material, ask questions, and take part in discussion or practice, they may be more likely to understand the material and see how it relates to their work.

The study does not call Learner Engagement a proven mediator. Formal mediation analysis would require a more demanding design and stronger assumptions about the temporal order of variables than the present cross-sectional survey can provide. Instead, engagement is treated as an intervening construct. The study tests whether it is associated with the antecedents and the training-effectiveness outcome.

Training Effectiveness

Training Effectiveness can be understood at different levels. Kirkpatrick's well-known framework distinguishes reaction, learning, behaviour, and results. The framework is useful because it shows why a positive immediate reaction is not sufficient evidence that a programme was effective. Employees may enjoy a session but not understand its content. They may understand the content but not have an opportunity to use it in their work. They may also apply a skill without being able to identify a wider organisation-level result.

Bahl, Kiran and Sharma (2024) provide a directly relevant banking study. Their open-access research used 402 responses from managerial and non-managerial employees of Indian public, private, and foreign-sector banks. Using Kirkpatrick's Reaction, Learning, Behaviour, and Results framework, the study evaluates employee training effectiveness in a banking setting. It does not examine Hybrid Learning, Trainer Competence, or Learner Engagement. Its value for the present study is to establish that employee training effectiveness is a substantive and measurable banking concern, while leaving the present four-construct combination unexamined.

Holton (1996) argues that evaluation levels do not fully explain why learning is or is not transferred into practice. Individual factors, training design, and the work environment may all influence later application. Baldwin and Ford (1988) similarly explain transfer of training as the generalisation and maintenance of learned capability in work situations. Blume et al. (2010) provide meta-analytic evidence that transfer is affected by individual, training-design, and work-environment factors.

These sources support a careful, employee-level definition for the present study. Training Effectiveness includes understanding what was taught, perceiving its relevance to the current role, feeling capable of using the learning, applying it to relevant tasks, dealing with related problems more effectively, and experiencing greater confidence in relevant responsibilities. The study does not ask employees to judge the Bank's overall productivity, customer satisfaction, reputation, or financial results. Those outcomes would require organisational data and a different research design.

Aziz (2015) developed a General Training Effectiveness Scale for Malaysian workplace learning. The study is useful because it treats learning and individual performance as distinct from organisation-level outcomes. The present questionnaire adopts the learning and individual-work logic while omitting organisation-level items. The resulting instrument is not presented as Aziz's scale or as a validated bank-specific scale. It is a short, context-adapted instrument whose content validity and reliability must be checked before interpretation.

Workplace implementation conditions

The evidence reviewed above suggests that hybrid learning should be considered as a learning design, not as a technology purchase. An LMS can make information and resources available, but it cannot decide which material employees need before a session, which issue requires discussion, or when learners need feedback. Similarly, a classroom session can allow discussion, but it may not give employees an easy way to revisit a policy, a procedure, or a short demonstration later. The quality of the connection between these activities is central to the present Hybrid Learning construct.

Mubayrik (2018) reviewed workplace blended-learning studies in adult education. The review includes studies from business, professional, and workplace settings and notes recurring themes such as flexible access, authentic learning activities, collaboration, learner satisfaction, and support for technology use. The review is useful for the present study because it shows that workplace blended learning has been investigated through varied designs, populations, and measures. It also shows why a single general claim about the effectiveness of hybrid learning is inadequate. The type of work, the learning objective, the technology, the trainer, and the opportunity to participate can all affect the employee experience.

The role of learning sequence is particularly relevant. In some programmes, digital material may be suitable before a trainer-led session. It can introduce terminology, policy content, background information, or a short demonstration. The face-to-face session can then focus on questions, case analysis, problem solving, or practice. In other programmes, a face-to-face session may introduce a complex issue and the platform may be used later for revision or a knowledge check. The present study does not test which sequence is best. Its Hybrid Learning items ask the employee whether the modes felt connected, whether the platform and materials were accessible, and whether material could be revisited.

For practical purposes, this connection can be understood as a sequence of digital preparation, trainer-led engagement and practice, workplace application, and digital reinforcement. The sequence is an illustrative learning-design principle, not a second theoretical model or a set of additional variables in the present study. It does not prescribe a fixed online-to-classroom ratio or one standard timetable. Its relevance lies in ensuring that each activity has a clear role and that employees can see how one activity supports the next.

The trainer's role also changes when learning is spread across digital and face-to-face activity. In a classroom-only session, the trainer may be responsible mainly for presentation, discussion, and immediate feedback. In a hybrid programme, the trainer may need to explain a pre-learning activity,

signal how a digital resource will be used later, respond to questions from different modes, and connect practice with the earlier content. Ansari et al. (2023), in a university setting, found that training lecturers in the planning, implementation, and assessment of blended learning improved their reported knowledge and skills in using blended-learning tools. The setting differs from the present employee survey, but it supports the practical view that facilitation across modes requires competence rather than merely access to technology.

These findings help distinguish Trainer Competence from Learner Engagement. A trainer can create clear opportunities for employees to participate, but the employee must still attend to the content, take part in an activity, and seek clarification. Conversely, an engaged employee may still struggle if the trainer does not explain the programme clearly or provide useful feedback. The two constructs are related but not interchangeable. Separating them improves the precision of the present model and avoids placing all responsibility for engagement on either the trainer or the employee.

The workplace setting also requires careful use of the term flexibility. Flexibility can mean that employees are able to complete material at a suitable time or revisit it when required. It does not necessarily mean that employees can study without time constraints or work pressure. Bank employees may have customer-facing duties, operational workloads, and fixed procedures that limit when learning can be completed. The present questionnaire therefore does not assume that flexibility is always available. It asks about access to materials and the ability to revisit digital content in the selected programme.

Union Bank's public reporting gives the study a specific LMS context. The 2023–24 annual report identifies Union Vidya as the Bank's LMS and describes integrated digital resources, including e-books, podcasts and audiobooks, social-learning platforms, microlearning modules, and virtual workshops. The 2024–25 report records e-learning, virtual training, staggered programmes, mandatory officer modules, role-based learning, and a Training Management System within Union Vidya (Union Bank of India, 2024, 2025). These reports support the relevance of an employee survey about hybrid learning. They do not provide evidence about employees' views of usability, access, integration, or trainer support in one particular programme. Those are empirical questions for the present survey.

Use of the literature in practical recommendations

The reviewed literature may also guide recommendations after data collection, but the recommendations should follow the actual findings. If employees report lower Hybrid Learning scores on access or material availability, the Bank may need to review the timing and accessibility of digital resources. If Trainer Competence scores are lower on explaining the connection between modes or giving feedback, trainer preparation may need attention. If Learner Engagement scores are lower on participation or clarification-seeking, programme design may need more opportunities for questions, practice, and follow-up.

Such recommendations should not be made in advance as conclusions. A cross-sectional survey will show how the selected employees perceive one recent programme. It will not show that a particular LMS feature, trainer intervention, or learning sequence caused a later improvement. The value of the literature is that it provides a reasoned framework for interpreting employee feedback and designing possible improvements after the results are known.

Summary of the empirical review

The reviewed literature supports several conclusions. Hybrid learning should be understood as a connected design, not simply the presence of two modes. Corporate evidence shows that training

effectiveness depends on more than whether content is delivered digitally or in a classroom; instructor involvement, design, participation, and practice matter. The wider blended-learning and engagement literature supports keeping Trainer Competence and Learner Engagement conceptually separate. Training-effectiveness literature shows why the survey should move beyond satisfaction while avoiding unsupported organisation-level claims.

At the same time, the literature is fragmented. Sosnova et al. provide a direct hybrid-learning and effectiveness proposition but use a student sample. Kim provides corporate training-effectiveness evidence but uses a comparative programme evaluation rather than an employee-perception path model. Bahl, Kiran and Sharma provide Indian banking evidence on training effectiveness through Kirkpatrick's model, but do not examine hybrid learning, trainer competence, or learner engagement. No single open source provides a complete ready-made model and questionnaire for the present Union Bank study. This justifies the use of three complementary base papers and separate measurement sources.

2.2 Theoretical Framework

The study uses three related perspectives: blended-learning integration, trainer-supported learner engagement, and training evaluation and transfer. These perspectives are not treated as three separate statistical models. Together, they explain why the four constructs are included.

Blended-learning integration

The first perspective is that the value of hybrid learning comes from purposeful integration. A digital activity may prepare employees for later discussion, while a face-to-face session can be used to explore complex points, demonstrate a procedure, and support practice. A later digital refresher may reinforce the learning. This perspective supports the Hybrid Learning construct, which includes connected activities, platform access, material availability, and revision opportunity.

The perspective does not require the same sequence for every programme. A compliance topic, a system update, and a customer-service topic may require different combinations of digital and face-to-face activity. The core principle is that employees should be able to understand the purpose and connection of the activities.

Trainer-supported learner engagement

The second perspective is that employees may be more likely to engage when trainers provide clear explanation, guidance, participation opportunities, and feedback. Kim identifies the importance of instructor engagement in the corporate hybrid-training setting. Ansari et al. and Mulaudzi provide additional evidence that facilitation and access conditions matter when learning is spread across digital and face-to-face activities. These sources support the rationale for the relationship hypotheses; they do not prove them in the present banking context.

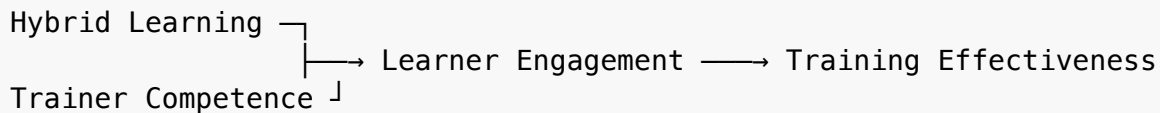
This perspective supports the two antecedent constructs in the present model. Hybrid Learning concerns the design and experience of the connected programme. Trainer Competence concerns the human support offered by the trainer. Learner Engagement concerns the employee's active involvement. Separating these concepts reduces the risk of treating every favourable opinion about a programme as evidence of the same variable.

Training evaluation and transfer

The third perspective is that effective training should support useful learning and work application. Kirkpatrick, Holton, Baldwin and Ford, and Blume et al. show why an evaluation should not rely only on completion or satisfaction. The present Training Effectiveness construct covers self-reported understanding, work relevance, capability, application, task support, and confidence. It remains limited to what an employee can reasonably report through one survey.

Conceptual framework

The theoretical perspectives support the following framework:



Hybrid Learning and Trainer Competence are expected to be positively related to Learner Engagement. Learner Engagement is expected to be positively related to Training Effectiveness. The arrows represent relationship hypotheses, not proof that one construct causes another. The study will use correlations and regression models that are appropriate to the cross-sectional, self-report design.

2.3 Case Study: Union Bank of India

The study is situated at Union Bank of India, with planned data collection among employees connected with Regional Office Chennai South. The Bank is a public-sector organisation operating in a context where employees need current knowledge of products, procedures, compliance requirements, risk controls, digital systems, and customer-service practices. Training is therefore relevant to both employee development and the consistent execution of work.

Union Bank's annual reports describe a broad Learning and Development system. The 2023–24 report records the launch of Union Vidya as the Bank's LMS and identifies digital resources and virtual workshops available through the platform. It also records classroom and locational training delivered through the wider training system. The 2024–25 report describes a Master Policy on Learning and Development that includes e-learning, virtual training, and staggered programmes. It reports ten mandatory e-learning modules for officers, role-based training, and the Training Management System in Union Vidya as a one-stop digital platform (Union Bank of India, 2024, 2025).

This documented infrastructure explains why Union Bank is an appropriate setting for the present study. It shows that the organisation has both digital-learning resources and trainer-led learning arrangements from which a connected hybrid programme may be designed. It does not establish that every employee has the same experience, every programme uses the same design, or digital resources are equally accessible across every setting. The survey addresses this employee-level question by asking eligible respondents about one specific recent programme that included both modes.

The case study is deliberately limited to actual employee experience. The questionnaire asks only eligible employees to select one recent programme that contained connected digital and face-to-face learning. This avoids asking respondents to judge the complete training system or to compare several programmes from memory. It also avoids collecting confidential information, customer information, system records, or performance data.

2.4 Gaps in the Study

The first gap is a **workplace-context gap**. Much of the blended-learning literature, including important environment scales, comes from higher education. Employee training involves different roles, schedules, and work pressures. Corporate studies exist, but direct evidence on how employees experience hybrid training remains less extensive than the education literature.

The second gap is a **banking-sector gap**. Bahl, Kiran and Sharma (2024) show that Training Effectiveness can be evaluated among Indian bank employees, but their Kirkpatrick-based study does not examine Hybrid Learning, Trainer Competence, Learner Engagement, or Union Vidya. The reviewed sources therefore do not provide a verified study of the present four-construct model among Union Bank of India employees. Banking employees need regular learning, and Union Bank has a documented digital learning platform, but the published organisational material does not show how employees experience the integration of digital and face-to-face learning, trainer support, engagement, and training effectiveness.

The third gap is a **mechanism gap**. Sosnova examines hybrid methods and learning effectiveness in education. Kim examines training effectiveness across delivery methods in a corporate setting. Bahl, Kiran and Sharma evaluate training effectiveness in Indian banking through Kirkpatrick's model. These studies are complementary, but none examines the present focused model in which Hybrid Learning and Trainer Competence are related to Learner Engagement and Learner Engagement is related to employee-level Training Effectiveness in a public-sector bank.

The fourth gap is a **measurement gap**. A validated blended-learning environment scale is available in a student setting, while a workplace training-effectiveness scale includes organisation-level outcomes. Neither can be used unchanged for a short survey of Union Bank employees. The present study develops a literature-informed instrument that separates programme design, trainer competence, learner engagement, and employee-level training effectiveness. The instrument is reviewed and piloted before the main survey.

The study does not claim that these gaps demonstrate an absence of all related research. They show that the evidence at the intersection of hybrid learning, trainer competence, learner engagement, training effectiveness, public-sector banking, and Union Vidya/LMS experience remains limited and fragmented. The study addresses that focused intersection.

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